INDEPENDENT STUDY PROJECT

Conceptual Design of a 1U CubeSat

A detailed, week-by-week delivery plan for an MSc Space Engineering portfolio

Compressed 6-Week Schedule · ~12–15 hrs/week

Prepared for: A BEng Mechanical Engineering graduate building space-systems evidence for postgraduate applications.
Mission baseline: 1U Earth-observation / environmental-monitoring satellite in a ~550 km Sun-synchronous orbit.
Outputs: 5–10 page technical report · CAD model · power/link/thermal budgets · trade-off study · 1-page portfolio summary.

00 · How to use this plan

This is a self-contained delivery plan. Everything you need to *actually finish* the project to a distinction standard is here — the mission baseline, the equations, the numbers to plug in, the weekly tasks, and the report structure. You are not expected to invent the physics; you are expected to **apply, justify, and interpret** it. That is exactly the skill MSc admissions tutors look for.

WHAT MAKES THIS "DISTINCTION LEVEL"

- **Quantified decisions.** Every subsystem choice is backed by a calculation, not a vibe.
- **Subsystem interaction.** You explicitly show how power limits comms, how orbit drives thermal, etc.
- **Honest constraints.** You state assumptions and admit limits. Examiners reward this heavily.
- **Traceability.** Requirements → design → trade-off → conclusion forms one clean thread.

RUNNING THIS ALONGSIDE THE ROCKET PROJECT

This plan assumes ~12–15 hrs/week on the CubeSat. The companion *Sounding Rocket Simulation* plan runs in parallel over the same six weeks. A combined weekly calendar is given in the appendix so the two never collide — CubeSat work is research/CAD/analysis heavy, the rocket is code heavy, so they balance well across a week.

> **WHY THIS BELONGS ON YOUR PERSONAL STATEMENT**
> A self-directed spacecraft conceptual design demonstrates exactly the competencies an MSc Space Engineering course assumes you can grow: systems thinking, subsystem literacy, constraint-driven design, and the ability to defend engineering decisions. Framing guidance is in Section 10.

· Contents

01 · Project overview & mission baseline

You will produce a **conceptual design** — not a build. The goal is a coherent paper spacecraft where every subsystem is sized with simple calculations and every decision is justified. To keep momentum, this plan fixes a concrete baseline mission. You may change it, but a fixed target prevents the most common failure mode: endless re-scoping.

Chosen baseline mission

Parameter	Baseline value	Why
Mission	Low-resolution Earth observation for environmental / vegetation & coastal monitoring	Achievable optical payload in 1U; rich justification material; clear data product.
Form factor	1U (100 × 100 × 113.5 mm with rails), mass ≤ 1.33 kg (up to 2 kg in recent CDS revisions)	Brief requirement; CDS-compliant.
Orbit	Sun-synchronous, ~550 km altitude, inclination ≈ 97.6°, ~95.5 min period, LTAN ~10:30	Consistent illumination & revisit for imaging; standard for EO.
Design life	1–2 years	Balances orbital decay, radiation dose, and COTS reliability.
Payload	COTS visible/NIR camera, ~50 m ground sample distance class	Honest 1U optical limit; suits regional environmental monitoring.

OPERATIONAL CONCEPT (CONOPS) IN ONE PARAGRAPH

The satellite is deployed from a P-POD into a dawn–dusk/morning SSO. It detumbles passively (or with magnetorquers), points nadir for imaging passes over target regions in daylight, stores images on-board, and downlinks during ground-station passes over the UHF amateur band. Power is collected by body-mounted solar cells, buffered in a Li-ion battery across eclipse, and the on-board computer manages a simple state machine: Detumble Safe Imaging Downlink Charge .

02 · Tools, setup & reference library

SOFTWARE (ALL FREE)

- **CAD:** Autodesk Fusion 360 (free for students), Onshape (browser, free), or FreeCAD.
- **Calcs:** A spreadsheet (Excel/Google Sheets) for the power, link & mass budgets — examiners love a clean budget table. Python optional for the orbit/thermal sweeps.
- **Diagrams:** draw.io / Excalidraw for the system block diagram and ConOps.
- **Writing:** Word or LaTeX (Overleaf). LaTeX signals seriousness but isn't required.

CORE REFERENCES

- **NASA "CubeSat 101"** — your single most useful document. Read it first.
- **CubeSat Design Specification (CDS), Cal Poly** — the dimensional/mass rules.
- **SMAD** — *Space Mission Analysis and Design* (Wertz/Larson) — the reference bible; dip in for budgets.
- Existing 1U missions for benchmarking (e.g. university CubeSat papers on power/link).

SET UP A PROJECT REPOSITORY ON DAY ONE

Create one folder (or a Git repo) with sub-folders: `/report`, `/cad`, `/calcs`, `/diagrams`, `/references`. A tidy structure you can screenshot or share is itself portfolio evidence of engineering discipline.

03 · Sequenced reading list

Read *just enough, just in time*. Don't try to read everything up front — match each reading to the week that needs it.

Week	Read this	So you can…
1	NASA CubeSat 101 (ch. 1–4); skim CDS	Write the mission statement & constraints list.
2	SMAD requirements chapter; one university CubeSat block diagram	Build the requirements table & system architecture.
3	"CubeSat power budget example" + LEO eclipse basics	Size solar array & battery.
4	"CubeSat thermal control basics"; ADCS intro (magnetorquers, gravity gradient)	Do the thermal balance & attitude concept.
5	UHF CubeSat link budget example; ground-station basics	Build the link budget & comms concept.
6	Review only — no new material	Write & polish.

04 · The 6-week schedule

Each week lists the focus, concrete tasks, and a hard deliverable. Treat the deliverable as non-negotiable — it is what becomes a report section. Total ≈ 75–90 hrs.

Week 1 — Mission & requirements

~12 hrs

Lock the mission so nothing downstream wobbles.

- Read NASA CubeSat 101 (ch. 1–4); note the 1U mass/volume rules from the CDS.
- Write a **mission statement** (2–3 sentences: who, what, why, where).
- Write the **ConOps** (the paragraph in §01, in your own words) and the operating-mode state machine.
- Draft a **constraints list**: mass ≤ 1.33 kg, volume 10 cm cube, deployable-free baseline, COTS-only, budget-free conceptual study, LEO.
- Start a top-level **requirements table** (functional + performance). Give each a tag (e.g. `PWR-01`).

Deliverable: 1-page mission summary + constraints list + first-draft requirements table.

Week 2 — Architecture & requirements flow-down

~13 hrs

Turn the mission into a system of interacting subsystems.

- Finalise the **requirements table** with quantified targets (orbit altitude, pointing accuracy, data rate, power, link margin, temperature range, lifetime).
- List the **subsystems**: Structure, EPS (power), ADCS, Comms, OBC/C&DH, Thermal, Payload.
- Draw the **system block diagram** showing power and data flows between subsystems (this single diagram earns "systems thinking" marks).
- Build a first **mass budget** table with margins (allocate per subsystem, keep ~20% reserve).
- Justify the **orbit**: compute period and velocity (see cheat-sheet §05.1); state why SSO at 550 km.

Deliverable: Requirements table + block diagram + mass budget + orbit justification (½ page calc).

Week 3 — Structure, layout & CAD + power budget

~15 hrs

Make it physical, then make it survive its energy balance.

- Build the **CAD model**: 1U frame, solar-cell faces, internal stack (PCBs as simple blocks), battery, payload aperture. Aim for a clean exploded view + an external render.
- Define a **layout** with a sensible centre-of-mass (matters for gravity-gradient stability).
- Compute the **power budget** (cheat-sheet §05.2): generation per illuminated face, orbit-average generation, load per mode, and the orbit energy balance across the eclipse.
- Confirm **positive energy margin**. If negative, iterate (reduce duty cycle, add a deployable panel, or lower data rate) — and *document the iteration*.

Deliverable: CAD model (2–3 screenshots/renders) + populated power budget table with margin.

Week 4 — Battery, thermal & attitude (ADCS)

~14 hrs

Close the energy storage loop; survive the temperature swing; point the satellite.

- **Battery sizing** (§05.3): eclipse energy demand → capacity at allowable depth-of-discharge. Check it fits in the 1U volume/mass.
- **Thermal balance** (§05.4): equilibrium temperature in sunlight and eclipse from absorbed solar vs radiated power; compare to component limits (battery −10…+45 °C is usually the tightest). Propose passive control (coatings, α/ε selection).
- **ADCS concept**: detumble + nadir pointing. Discuss passive (gravity gradient, permanent magnet + hysteresis rods) vs active (magnetorquers, reaction wheels) and pick one with justification. State the pointing requirement the payload imposes.

Deliverable: Battery sizing calc + thermal balance (with numbers) + ½-page ADCS concept & justification.

Week 5 — Communications, payload & link budget

~14 hrs

Get the data home and prove there's enough margin to do it.

- **Comms concept:** choose band (UHF ~437 MHz amateur is the realistic 1U default), modulation (GMSK), data rate (e.g. 9.6 kbps), antenna (deployable dipole/monopole).
- **Link budget** (§05.5): EIRP, free-space path loss at max range, ground G/T, required Eb/N0, and the resulting **link margin** (target ≥ 3 dB).
- **Payload & data:** compute ground sample distance (§05.6) and data volume per image; compare to downlink capacity per pass — a perfect "subsystem interaction" discussion.
- Reconcile: does comms power fit inside the Week 3 power budget? Iterate if not.

Deliverable: Link budget table (≥3 dB margin) + GSD/data-volume calc + comms concept.

Week 6 — Trade-offs, integration & write-up

~16 hrs

Tie it together, show the trade-offs, and produce the deliverables.

- Build the **trade-off study** (§06): e.g. altitude (resolution vs lifetime vs radiation), comms band, ADCS approach, body-mounted vs deployable panels. Use a weighted decision matrix.
- Do a **systems integration review**: one table showing every requirement and whether the design meets it (compliance matrix).
- Write the **full report** (§07 structure), insert CAD renders, budgets and the block diagram.
- Produce the **1-page application summary** (§10).
- Final pass: assumptions stated, limitations discussed, figures labelled, references cited.

Deliverable: Final 5–10 page report + trade-off table + compliance matrix + 1-page portfolio summary.

05 · Engineering cheat-sheet

Every number you need to size this satellite. Constants: μ_E = 398 600 km³/s², R_E = 6371 km, solar constant G_s = 1361 W/m², σ = 5.67×10^{-8} W/m²K⁴.

05.1 Orbit (period, velocity, eclipse)

For a circular orbit at altitude *h*, with orbital radius *r* = R_E + *h*:

$$v = \sqrt{\mu_E / r} \qquad T = 2\pi \cdot \sqrt{r^3 / \mu_E}$$

Worked (h = 550 km): r = 6921 km → v = √(398600/6921) ≈ **7.59 km/s**; T = 2π·√(6921³/398600) ≈ 5731 s ≈ **95.5 min** (~15.1 orbits/day). Take the eclipse fraction for LEO as ~35% → eclipse ≈ **33 min**, sunlight ≈ 62 min (state this as a worst-case assumption; the rigorous version uses the β-angle and Earth's angular radius).

05.2 Power budget

Generation from one fully-illuminated body-mounted face of area A_{face} with cell efficiency η and packing factor p:

$$P_{face} = G_s \cdot A_{face} \cdot \eta \cdot p \cdot \cos\theta$$

Worked: A_{face}=0.01 m², η=0.30 (triple-junction), p≈0.8, normal incidence → P_{face} ≈ 1361×0.01×0.30×0.8 ≈ **3.3 W** peak. Because the body tumbles/points and only some faces are lit at favourable angles, take an **orbit-average generation of ~1.5–2 W** for body-mounted 1U (justify your figure). Orbit energy balance over one period:

$$E_{gen} = P_{avg,gen} \cdot t_{sun} \qquad E_{load} = \Sigma\,(P_{mode} \cdot t_{mode}) / \eta_{dist}$$

Tabulate load per mode (OBC idle ~0.2 W, comms TX ~2–4 W for a few minutes/pass, ADCS ~0.2 W, payload during imaging). **Require $E_{gen} \geq E_{load}$ over the orbit** with margin; if not, iterate.

05.3 Battery sizing

$$C_{batt}\ [\mathrm{Wh}] = (P_{eclipse} \cdot t_{eclipse}) / (DoD \cdot \eta_{batt})$$

Worked: $P_{eclipse}$ ≈ 2 W, $t_{eclipse}$ ≈ 0.55 h, DoD = 0.25 (long cycle life), η_{batt} = 0.9 → C ≈ (2×0.55)/(0.25×0.9) ≈ **4.9 Wh**. A typical 1U battery is 8–20 Wh, so this fits comfortably with healthy margin.

05.4 Thermal balance

Steady-state: absorbed solar = radiated. α = absorptivity, ε = emissivity, A_{sun} = sunlit projected area (~0.01 m² for one face), A_{rad} = total radiating area (0.06 m² for a cube):

$$\alpha \cdot G_s \cdot A_{sun} = \varepsilon \cdot \sigma \cdot A_{rad} \cdot T^4 \rightarrow T = [\,\alpha\, G_s\, A_{sun} / (\varepsilon\, \sigma\, A_{rad})\,]^{1/4}$$

Worked (α=0.7, ε=0.8): T = [0.7·1361·0.01 / (0.8·5.67e-8·0.06)]$^{1/4}$ ≈ **243 K ≈ −30 °C** sunlit average; eclipse (no solar) is colder. Use this to argue for surface coatings / α-ε tuning and to check the battery stays in range. Add Earth albedo + IR as a refinement and note them as a limitation if omitted.

05.5 Communications link budget

Free-space path loss at slant range *d* (km) and frequency *f* (MHz):

$$\mathrm{FSPL}\ [\mathrm{dB}] = 32.45 + 20\cdot\log_{10}(d) + 20\cdot\log_{10}(f) \quad \mathrm{Margin} = \mathrm{EIRP} + G_{rx} - \mathrm{FSPL} - L_{misc} - (E_b/N_0)_{req} - 10\log_{10}(R) - k - T_{sys}$$

Practical approach: build a line-by-line table (TX power, antenna gains, FSPL, pointing/polarisation/ atmos losses, system noise temperature, required Eb/N0 for your modulation+coding, data rate). At 437 MHz, d≈2000 km (low elevation): FSPL ≈ 32.45 + 66 + 52.8 ≈ **151 dB**. Aim for a final **link margin ≥ 3 dB**. If short, reduce data rate, raise TX power, or add ground-station gain — and document the choice.

05.6 Payload: ground sample distance & data volume

$$\mathrm{GSD} = (p_{pix} \cdot H) / f \qquad \mathrm{Data/image} = N_{pix} \cdot \mathrm{bits/pixel}$$

Worked: pixel pitch p=3.45 µm, focal length f=35 mm, H=550 km → GSD = (3.45e-6×550e3)/0.035 ≈ **54 m/pixel** — honest for a 1U, suitable for regional environmental monitoring. Then compare image size to downlink per pass (e.g. 9.6 kbps × ~6 min usable ≈ 3.5 Mbit ≈ 0.43 MB/pass) to motivate on-board compression / selective downlink — a strong systems-interaction point.

06 · Trade-off study framework

Systems thinking is 25% of the marks. Use a **weighted decision matrix**: list options as rows, criteria as columns, score each 1–5, multiply by a weight, sum. Show your reasoning, not just the winner.

Trade to study	Competing criteria	The tension you must articulate
Orbit altitude (450 vs 550 vs 700 km)	Resolution, lifetime, radiation, comms range	Lower = sharper images + shorter life (drag); higher = longer life + coarser images + more radiation.
Solar: body-mounted vs deployable	Power, complexity, mass, risk	Deployables ~double power but add a deployment failure mode and mass.
ADCS: passive vs active	Pointing accuracy, power, cost, mass	Passive is cheap/robust but coarse; active hits payload pointing needs at a power/complexity cost.
Comms band: UHF vs S-band	Data rate, hardware availability, ground segment	S-band gives far higher rates but heavier/pricier radios and harder pointing.

MAKE IT A THREAD, NOT A TABLE DUMP

The best reports carry a single trade — usually altitude — through the whole document, showing how it ripples into power, thermal, comms and lifetime. That demonstrates the interaction examiners are scoring.

07 · Report structure mapped to the marking criteria

Report section	Feeds which criterion	Weight
1. Introduction & mission statement	Mission & requirements clarity	15%
2. ConOps & operating modes	Mission clarity / systems thinking	—
3. Requirements table & constraints	Mission & requirements clarity	15%
4. System architecture & block diagram	Systems thinking	25%
5. Subsystem design (Structure, EPS, Thermal, ADCS, Comms, Payload) with calcs	Subsystem understanding + engineering reasoning	20% + 25%
6. Orbit selection & justification	Engineering reasoning	25%
7. Trade-off study & compliance matrix	Systems thinking	25%
8. Assumptions, limitations & future work	Engineering reasoning	25%
9. Conclusion + CAD figures throughout	Presentation	15%

Keep it to 5–10 pages of body. Budgets and the compliance matrix can be appendices. Number every figure and table and refer to each in the text.

08 · Distinction tips & common pitfalls

DO

- State every assumption inline and revisit it in the limitations section.
- Show the iteration: "first pass gave negative margin, so I...".
- Keep one consistent set of numbers across all budgets.
- Benchmark against a real 1U mission to sanity-check your figures.
- Label units everywhere; carry margins on mass, power and link.

AVOID

- Quoting power/thermal numbers with no calculation behind them.
- Designing subsystems in isolation with no interaction story.
- Over-claiming resolution/data rate beyond 1U reality.
- A CAD model with no internal layout or centre-of-mass thought.
- Forgetting the eclipse — it drives battery and thermal.

09 · Combined parallel calendar

Suggested split when running this and the rocket simulation together (~25–28 hrs/week total). The two are complementary: CubeSat weeks are research/CAD/analysis; rocket weeks are coding.

Week	CubeSat focus (≈12–15 hr)	Rocket focus (≈12–14 hr)
1	Mission, ConOps, constraints	Python refresh + forces reading
2	Requirements, block diagram, orbit calc	Vertical thrust-vs-gravity model
3	CAD + power budget	Add drag (quadratic)
4	Battery, thermal, ADCS	Variable mass + RK4 + verification
5	Comms, payload, link budget	Trade studies + figures
6	Trade-offs, integration, write-up	Report + portfolio summary

10 · Personal-statement framing

A MODEL PARAGRAPH YOU CAN ADAPT (DON'T COPY VERBATIM)

"To deepen my preparation for postgraduate space engineering, I undertook a self-directed conceptual design of a 1U Earth-observation CubeSat. Working from the CubeSat Design Specification, I defined a Sun-synchronous mission, flowed it down into a requirements set, and sized each subsystem with first-order analysis — closing a power budget across the orbital eclipse, sizing the battery to a 25% depth-of-discharge, and verifying a positive UHF link margin. I carried a single altitude trade-off through the design to show how it propagated into resolution, lifetime and thermal balance, and documented every assumption and limitation. The project taught me to reason about a spacecraft as a system of competing constraints rather than isolated parts."

ONE-PAGE PORTFOLIO SUMMARY — INCLUDE

- A render of the CAD model + the system block diagram (visual hook).

- Mission one-liner, orbit, key budgets (mass / power / link margin) as headline figures.
- The headline trade-off and your conclusion.
- Two sentences on what you'd do next (deployable panels, higher-fidelity thermal model) — shows self-critique.

FINAL NOTE

A strong submission explains decisions, shows awareness of constraints, demonstrates subsystem interaction, and presents a coherent spacecraft. Every week above is built to produce exactly that evidence.